ANSWER KEY & MARKING SCHEME SET A

Chapter 9: **Life Processes in Plants** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** photosynthesis	uses sunlight
2	**(d)** guard cells	control opening
3	**(c)** starch	iodine test
4	**(b)** xylem	from roots
5	**(d)** pulling water upward and cooling the plant	suction pull
6	**(b)** pitcher plant	traps insects for nitrogen
7	**(b)** nitrogen	poor soil
8	**(d)** guard cells	control gas exchange
9	**(d)** parasitic	host harmed
10	**(c)** the stalk joining leaf to stem	part of leaf

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Process by which green plants make food using sunlight, water and CO_2 (2).	2
12	Carbon dioxide + Water → Glucose + Oxygen, in sunlight and chlorophyll (2).	2
13	Tiny pores on leaves (1); exchange gases and release water vapour (1).	1+1
14	Traps sunlight energy for photosynthesis (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Flat, thin blade with large area (1); chlorophyll in cells (1); stomata and veins (1).	1+1+1
16	Guard cells open the pore for gas exchange (1.5); close to reduce water loss (1.5).	1½+1½
17	Cover part of a leaf with black paper (1); expose to sun, remove chlorophyll and add iodine (1); covered part shows no starch (1).	1+1+1
	OR: Makes food for all (1.5); releases oxygen (1.5).	1½+1½
18	Root hairs absorb (1); xylem carries upward (1); transpiration pulls (1).	1+1+1
	OR: Leaves make food (1); phloem carries to all parts (1); stored in roots, fruits (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Making food using sunlight (1). (ii) CO_2, water → glucose, O_2 (2). (iii) Chloroplasts in leaves (2).	1+2+2
	OR: (i) Role of light (1). (ii) To see iodine colour (1). (iii) Covered part no blue-black (1). (iv) Light is necessary (2).	1+1+1+2
20	(i) Pores (2). (ii) Open and close (2). (iii) Gas exchange and transpiration (1).	2+2+1
	OR: (i) A (1). (ii) Lack of water (1). (iii) No light (1). (iv) Both needed (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET B

Chapter 9: **Life Processes in Plants** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** chlorophyll	traps light energy
2	**(b)** oxygen	by-product
3	**(d)** starch	test for starch
4	**(a)** phloem	translocation
5	**(b)** all day and night	all living cells
6	**(a)** saprotrophs	e.g. mushrooms
7	**(a)** darker green and receives more light	palisade layer
8	**(c)** green plants	make own food
9	**(b)** pale and weak	no photosynthesis
10	**(c)** transport of water and food	contain xylem and phloem

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Xylem carries water and minerals (1); phloem carries food (1).	1+1
12	Loss of water vapour from leaves through stomata (2).	2
13	Pulls water upward and cools the plant (2).	2
14	Remove chlorophyll in warm alcohol (1); add iodine, blue-black shows starch (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Xylem: water, minerals, upward (1.5); phloem: food, both directions (1.5).	1½+1½
16	Photosynthesis stores energy, uses CO_2, in light (1.5); respiration releases energy, uses O_2, always (1.5).	1½+1½
17	Parasitic (1); insectivorous (1); saprotrophic (1).	1+1+1
	OR: Only green patches have chlorophyll (1.5); only they turn blue-black with iodine (1.5).	1½+1½
18	Rhizobium fixes nitrogen (1.5); enriches soil (1.5).	1½+1½
	OR: Pitcher-shaped leaf with slippery rim (1.5); insects fall and are digested (1.5).	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Roots to leaves via xylem (2). (ii) Phloem (2). (iii) Xylem and phloem (1).	2+2+1
	OR: (i) Water loss (1). (ii) Heat, wind, humidity (2). (iii) Cooling, pull (2).	1+2+2
20	(i) Cuscuta (1). (ii) Parasitic (1). (iii) Weakens host (1). (iv) Insectivorous or saprotrophic (2).	1+1+1+2
	OR: (i) Glucose + O_2 → CO_2 + water + energy (1). (ii) Opposites (2). (iii) Needs energy always (2).	1+2+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.